

Quantum State – Information – Causality – Spacetime

A computational research ansatz for time-independent relational dynamics

Research ansatz – not an established theory of quantum gravity or time travel.

1. Research question

The computational problem is

Can a physical theory containing quantum state, information and causal structure reproduce a family of spacetime histories that standard QM+GR normally describe using an external classical time parameter?

 (1)

The central design principle is to avoid assuming that the parameter t is fundamental. Instead, the model attempts to obtain temporal ordering relationally from quantum state, information and causal structure.

The four principal objects are

$\rho \longleftrightarrow I \longleftrightarrow C \longleftrightarrow g_{\mu\nu}$

 (2)

where

- ρ is the quantum state,
- I is the information/correlation structure,
- C is causal influence/order,
- $g_{\mu\nu}$ is spacetime geometry.

The model must reduce to established physics when the proposed new sectors are turned off.

2. Fundamental Hilbert space

For a toy model, take

$$\mathcal{H} = \mathcal{H}_M \otimes \mathcal{H}_C \otimes \mathcal{H}_O, \quad (3)$$

where

- $\mathcal{H}_M$: matter degrees of freedom,

- $\mathcal{H}_C$: internal clock/reference degrees of freedom,
- $\mathcal{H}_O$: observer/memory degrees of freedom.

The global physical state is

$$\rho_{\text{phys}} \in \mathcal{D}(\mathcal{H}), \quad \rho_{\text{phys}} \geq 0, \quad \text{Tr}(\rho_{\text{phys}}) = 1. \quad (4)$$

No external t is assumed at this level.

3. Hamiltonian constraint instead of external time

Replace the ordinary Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad (5)$$

with a global physical-state constraint

$$\boxed{\hat{\mathcal{C}} |\Psi\rangle = 0.} \quad (6)$$

For the exploratory model,

$$\hat{\mathcal{C}} = \hat{H}_M + \hat{H}_C + \hat{H}_O + \hat{H}_{\text{int}} + \hat{\mathcal{Q}}. \quad (7)$$

Here $\hat{\mathcal{Q}}$ is an explicitly unknown extension sector.

The baseline is

$$\boxed{\hat{\mathcal{Q}} = 0} \quad (8)$$

and must reproduce ordinary quantum mechanics in the appropriate regime.

4. Relational time

Let $\hat{T}$ be an internal clock observable and let Π_τ be the projector associated with clock reading τ .

The conditional matter state is

$$\boxed{\rho_M(\tau) = \frac{\text{Tr}_{C,O} [(\Pi_\tau \otimes I) \rho_{\text{phys}}]}{p(\tau)}} \quad (9)$$

with

$$p(\tau) = \text{Tr} [(\Pi_\tau \otimes I) \rho_{\text{phys}}]. \quad (10)$$

Thus an effective history

$$\tau \mapsto \rho_M(\tau) \quad (11)$$

can emerge from correlations with an internal clock rather than from an externally imposed classical time variable.

5. Information structure

For a subsystem A ,

$$S(A) = -\text{Tr}(\rho_A \log \rho_A) \quad (12)$$

is the von Neumann entropy.

For two subsystems A, B ,

$$\boxed{I(A : B) = S(A) + S(B) - S(AB)}. \quad (13)$$

For a discrete collection of subsystems,

$$\mathbf{I}_{ij} = I(i : j). \quad (14)$$

This produces an information/correlation network without yet interpreting it as geometry.

The information is explicitly not assumed to be spacetime distance.

6. Causal structure

Instead of imposing a binary causal graph, define an operational causal influence matrix

$$\boxed{C_{ij} \in [0, 1]}. \quad (15)$$

One candidate definition is

$$C_{ij} = \sup_{X_i, X'_i} D\left(\rho_{j|X_i} \parallel \rho_{j|X'_i}\right), \quad (16)$$

where X_i, X'_i are alternative interventions at subsystem i , and

$$D(\rho \parallel \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)] \quad (17)$$

is quantum relative entropy.

The interpretation is

$$C_{ij} = 0 \quad \Rightarrow \quad \text{no detected operational influence } i \rightarrow j, \quad (18)$$

while $C_{ij} > 0$ indicates the ability of an intervention at i to alter the state observed at j .

7. General quantum process

To avoid assuming a fixed causal order from the start, introduce a general process object

$$\boxed{W} \tag{19}$$

and local operations $M_i^{a_i}$.

The probabilities of observed outcomes are

$$\boxed{P(a_1, \dots, a_n) = \text{Tr}[W (M_1^{a_1} \otimes \dots \otimes M_n^{a_n})]}. \tag{20}$$

The computational model should permit both ordinary fixed causal order and, where mathematically consistent, indefinite causal order.

8. Information-induced distance: exploratory ansatz

Do not assume that mutual information *is* a metric.

Instead define a testable ansatz.

First define

$$w_{ij} = \frac{I_{ij} + \varepsilon}{I_{\max} + \varepsilon}, \tag{21}$$

then define

$$\boxed{d_I(i, j) = -\ell \log w_{ij}}. \tag{22}$$

This is deliberately only a computational hypothesis.

More generally, allow

$$\boxed{d_{\text{trial}}(i, j) = F_{\theta}(I_{ij}, C_{ij})} \tag{23}$$

with a constrained function F_{θ} .

A low-complexity basis may be

$$\begin{aligned} F_{\theta}(I, C) = & a_0 + a_1[-\log(I + \varepsilon)] + a_2C + a_3C^2 \\ & + a_4IC + \dots \end{aligned} \tag{24}$$

The model must learn or optimize θ under physical constraints rather than merely fitting arbitrary distances.

9. Reconstructing spacetime geometry

Associate each event/subsystem i with a coordinate representation

$$x_i^\mu \in \mathbb{R}^{1+d}. \quad (25)$$

Let

$$\Delta x_{ij}^\mu = x_i^\mu - x_j^\mu. \quad (26)$$

For a candidate Lorentzian metric $g_{\mu\nu}$,

$$\boxed{s_{ij} = g_{\mu\nu}(x) \Delta x_{ij}^\mu \Delta x_{ij}^\nu.} \quad (27)$$

The model attempts to find $g_{\mu\nu}$ and x_i^μ such that

$$d_g(i, j) \approx F_\theta(I_{ij}, C_{ij}). \quad (28)$$

This does not assert that information generates spacetime. It asks whether a geometry can be reconstructed consistently from the available relational data.

10. Causal compatibility

For signature

$$(-, +, +, +), \quad (29)$$

a timelike or null separation satisfies

$$s_{ij} \leq 0. \quad (30)$$

For a directed causal influence $i \rightarrow j$, the candidate embedding should satisfy

$$C_{ij} > 0 \implies s_{ij} \leq 0, \quad (31)$$

together with the ordering constraint

$$x_j^0 > x_i^0 \quad (32)$$

when a single-valued temporal ordering exists.

The model must *not* hard-code acyclicity if the process formalism admits a nontrivial closed or indefinite causal structure.

11. General relativity baseline

For the classical limit,

$$\boxed{G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.} \quad (33)$$

Define the Einstein residual

$$R_{\mu\nu}^{GR} = G_{\mu\nu} + \Lambda g_{\mu\nu} - \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (34)$$

The baseline loss is

$$L_G = \left\| R_{\mu\nu}^{GR} \right\|^2. \quad (35)$$

12. Unknown extension sector

Introduce a deliberately agnostic correction

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} + \Delta_{\mu\nu}. \quad (36)$$

The correction is constrained to depend on the relational/quantum structure:

$$\Delta_{\mu\nu} = \Delta_{\mu\nu}[\rho, W, I, C]. \quad (37)$$

The classical theory must remain a valid limit:

$$\Delta_{\mu\nu} \rightarrow 0 \quad (38)$$

in every experimentally validated GR regime.

13. Conservation constraint

Because the Einstein tensor obeys the contracted Bianchi identity,

$$\nabla^\mu G_{\mu\nu} = 0, \quad (39)$$

a consistent extension should satisfy

$$\nabla^\mu T_{\mu\nu} = 0 \quad (40)$$

and therefore

$$\nabla^\mu \Delta_{\mu\nu} = 0. \quad (41)$$

This prevents the model from obtaining a numerical “solution” by silently creating or destroying stress-energy.

14. Parametric correction sector

Let

$$\Delta_{\mu\nu} = \sum_k \theta_k B_{\mu\nu}^{(k)}. \quad (42)$$

Candidate basis tensors can be built from information and causal scalar fields ϕ_I, ϕ_C , for example

$$B_{\mu\nu}^{(1)} = g_{\mu\nu} \phi_I, \quad (43)$$

$$B_{\mu\nu}^{(2)} = \nabla_\mu \nabla_\nu \phi_I, \quad (44)$$

$$B_{\mu\nu}^{(3)} = \nabla_\mu \phi_I \nabla_\nu \phi_I, \quad (45)$$

$$B_{\mu\nu}^{(4)} = g_{\mu\nu} (\nabla \phi_I)^2, \quad (46)$$

with analogous terms constructed from ϕ_C , and mixed terms where allowed.

The coefficients θ_k are not accepted merely because they reduce a loss. They must survive all physical consistency tests.

15. Effective action

A compatible effective-action representation is

$$\boxed{S = S_{\text{EH}} + S_M + S_{\text{info}} + S_{\text{causal}} + S_{\text{coupling}}.} \quad (47)$$

The Einstein–Hilbert term is

$$S_{\text{EH}} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda). \quad (48)$$

An exploratory coupling sector can be written as

$$S_{\text{coupling}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{coupling}}(\phi_I, \phi_C, g_{\mu\nu}, \rho, W). \quad (49)$$

The equations of motion follow from

$$\boxed{\frac{\delta S}{\delta g^{\mu\nu}} = 0} \quad (50)$$

and the corresponding variations with respect to matter, clock, and information variables.

16. Time-independent history representation

Instead of integrating an equation in external time, represent a candidate history as

$$\boxed{\Gamma = \{(\rho_i, W_i, C_i, g_i)\}_{i=1}^N.} \quad (51)$$

The index i is a computational label, not an assumed physical time.

A relational temporal parameter τ_i is then inferred from causal and clock correlations:

$$C_{ij} > 0 \quad \Rightarrow \quad \tau_i < \tau_j \quad (52)$$

whenever an ordinary causal ordering is admissible.

The question is therefore whether

$$\Gamma \rightarrow \tau \quad (53)$$

can emerge without externally imposing t .

17. State-to-state transformation

Given two physical states

$$\rho_A, \quad \rho_B, \quad (54)$$

define a quantum channel

$$\boxed{\mathcal{R}_{A \rightarrow B} : \rho_A \mapsto \rho_B.} \quad (55)$$

The map must be completely positive and trace preserving:

$$\mathcal{R}(\rho) \geq 0, \quad \text{Tr}[\mathcal{R}(\rho)] = 1. \quad (56)$$

The state reconstruction fidelity is

$$\boxed{F(\rho_B, \mathcal{R}_{A \rightarrow B}[\rho_A]).} \quad (57)$$

A perfect reconstruction would require

$$F \rightarrow 1. \quad (58)$$

This alone is *not* time travel.

18. Reversibility test

Define

$$L_{\text{state}} = 1 - F(\rho_A, \mathcal{R}_{B \rightarrow A}[\rho_B]). \quad (59)$$

For a round trip,

$$L_{\text{round}} = D(\rho_B, (\mathcal{R}_{A \rightarrow B} \circ \mathcal{R}_{B \rightarrow A})(\rho_B)). \quad (60)$$

A vanishing round-trip loss means the transformation is reversible within the model. It does not by itself imply temporal reversal.

19. Closed causal self-consistency

Let

$$\mathcal{E}_\Gamma \tag{61}$$

be the generalized process induced by the candidate causal/spacetime structure.

A closed self-consistent configuration satisfies

$$\boxed{\rho^\star = \mathcal{E}_\Gamma(\rho^\star)}. \tag{62}$$

The search is therefore for a nontrivial fixed point subject simultaneously to quantum, information, causal and geometric constraints.

A candidate solution must satisfy, at minimum,

$$\hat{\mathcal{C}}\rho^\star = 0, \tag{63}$$

$$\nabla^\mu \Delta_{\mu\nu} = 0, \tag{64}$$

$$L_G \approx 0, \tag{65}$$

$$L_I \approx 0, \tag{66}$$

$$L_C \approx 0. \tag{67}$$

20. Historical target state

For a concrete target such as the local environment of Baba Novac railway station in 1987, define a target class of observables

$$\boxed{\mathcal{T}_{1987} = \{O_1, O_2, \dots, O_K\}}. \tag{68}$$

Possible observables in a toy representation include

$$O_1 : \text{rail geometry}, \tag{69}$$

$$O_2 : \text{train position/state}, \tag{70}$$

$$O_3 : \text{signal state}, \tag{71}$$

$$O_4 : \text{environmental state}, \tag{72}$$

$$O_5 : \text{biological/observer state}. \tag{73}$$

The target loss is

$$\boxed{L_{\text{target}} = \sum_k w_k D \left[O_k(\rho, g, C), O_k^{1987} \right]}. \tag{74}$$

The historical year is therefore not treated as a fundamental dynamical variable. It specifies a target class of physical observables and relations.

21. Global optimization problem

Collect the model variables:

$$\Theta = \{\rho, W, C, x_i^\mu, g_{\mu\nu}, \theta, \mathcal{Q}, \Delta\}. \quad (75)$$

Define

$$\mathcal{L} = \lambda_Q L_Q + \lambda_I L_I + \lambda_C L_C + \lambda_G L_G + \lambda_E L_E + \lambda_{\text{rel}} L_{\text{rel}} + \lambda_{\text{target}} L_{\text{target}} + \lambda_{\text{new}} L_{\text{new}}. \quad (76)$$

Representative terms are

$$L_I = \sum_{i < j} \omega_{ij} [d_g(i, j) - F_\theta(I_{ij}, C_{ij})]^2, \quad (77)$$

$$L_E = \|\nabla^\mu T_{\mu\nu} + \nabla^\mu \Delta_{\mu\nu}\|^2, \quad (78)$$

and

$$L_{\text{new}} = \|\mathcal{Q}\|^2 + \|\Delta\|^2. \quad (79)$$

The full computational task is

$$\min_{\Theta} \mathcal{L}(\Theta) \quad (80)$$

subject to positivity, normalization, complete positivity, conservation, symmetry, covariance and known classical/quantum limits.

22. The decisive baseline comparison

Three model classes should be compared.

Model A: Standard

$$\text{QM} + \text{GR} + t_{\text{external}}. \quad (81)$$

Model B: Relational time

$$\text{QM} + \text{GR} + \text{internal clock}. \quad (82)$$

Model C: Extended relational framework

$$\text{QM} + \text{Information} + \text{Causality} + \text{Geometry} + (\mathcal{Q}, \Delta). \quad (83)$$

The extension is only accepted if it produces a genuine improvement that cannot be reproduced by Model A or B while respecting all physical constraints.

23. Falsification requirements

A candidate extension is rejected if its apparent success is attributable to

- numerical discretization,
- finite-size effects,
- coordinate choice,
- overfitting,
- an arbitrary information-to-distance ansatz,
- a hidden external time variable,
- violation of energy-momentum conservation,
- violation of quantum positivity or complete positivity,
- violation of established causal constraints,
- a known QM or GR effect being misidentified as new physics.

A viable extension should generate at least one observable prediction

$$\boxed{\mathcal{O}_{\text{new}} \neq \mathcal{O}_{\text{QM+GR}}} \quad (84)$$

for some experimentally accessible regime, with the difference surviving reasonable changes of numerical implementation.

24. What would count as an actual result?

The desired result is *not*

$$\text{“the simulation produced a time machine.”} \quad (85)$$

The scientifically meaningful sequence is

$$\boxed{\text{known physics} \rightarrow \text{identified inconsistency} \rightarrow \text{minimal extension} \rightarrow \text{new prediction} \rightarrow \text{falsifiable experiment.}} \quad (86)$$

A particularly interesting outcome would be a consistent solution in which

$$\boxed{\text{no external } t} \quad (87)$$

is required, while quantum state, information and causal structure jointly generate a spacetime history satisfying the known classical limits.

An even stronger result would be a nontrivial self-consistent solution

$$\boxed{\rho^* = \mathcal{E}_\Gamma(\rho^*)} \quad (88)$$

with a causal structure that cannot be reduced to an ordinary acyclic classical time ordering.

Such a result would still require an independent physical interpretation and experimental validation. It would not, by itself, establish physical time travel.

25. Core conceptual distinction

The model explicitly distinguishes four operations:

$$\text{state reconstruction : } \rho_B \rightarrow \rho_A, \quad (89)$$

$$\text{simulation : } \text{produce observations indistinguishable from } A, \quad (90)$$

$$\text{resurrection/reconstruction of an observer : } \mathcal{H}_O^{(B)} \rightarrow \mathcal{H}_O^{(A)}, \quad (91)$$

$$\text{genuine temporal travel : } \text{a causal/worldline structure connecting the observer to an earlier} \quad (92)$$

The computational framework must never silently identify these four concepts.

26. Final compact formulation

The entire research program can be summarized as

Find $(\rho, W, C, g, \mathcal{Q}, \Delta)$
 such that
 $\hat{C}\rho = 0, \quad \rho \geq 0, \quad \text{Tr } \rho = 1,$
 $P(a_1, \dots, a_n) = \text{Tr} \left[W \bigotimes_i M_i^{a_i} \right],$
 $d_g(i, j) \approx F_\theta(I_{ij}, C_{ij}),$
 $G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} + \Delta_{\mu\nu},$
 $\nabla^\mu \Delta_{\mu\nu} = 0,$
 $\Delta_{\mu\nu} \rightarrow 0, \quad \mathcal{Q} \rightarrow 0 \quad \text{in known physical limits},$
 $\rho^* = \mathcal{E}_\Gamma(\rho^*),$
 $L_{\text{target}} \rightarrow 0,$

(93)

while minimizing

$$\mathcal{L} = \lambda_Q L_Q + \lambda_I L_I + \lambda_C L_C + \lambda_G L_G + \lambda_E L_E + \lambda_{\text{rel}} L_{\text{rel}} + \lambda_{\text{target}} L_{\text{target}} + \lambda_{\text{new}} \left(\|\mathcal{Q}\|^2 + \|\Delta\|^2 \right).$$

(94)

The crucial baseline is

$$\mathcal{Q} = \Delta_{\mu\nu} = 0.$$

(95)

Only a demonstrable failure of the baseline, followed by a constrained and falsifiable extension, constitutes evidence that something beyond the current formalism is required.